

Lem nucleo dirichlet no es nucleo sumabilidad

Lema 1. $\forall N \in \mathbb{N} : \int_{-1/2}^{1/2} |D_N(x)| dx \geq C \log(N)$ para alguna constante $C > 0$.

Demostración: Por el `lem-formula-nucleo-dirichlet/??`, sabemos que D_N es una función par y sus ceros en $[0, 1/2)$ son los $x \in \mathbb{R}$ tales que $(2N+1)\pi x = k\pi \iff x = \frac{k}{2N+1}$ para $k = \mathbb{N}_N$. Por tanto,

$$\begin{aligned}
 \int_{-1/2}^{1/2} |D_N(x)| dx &\stackrel{\text{par}}{=} 2 \int_0^{1/2} |D_N(x)| dx = 2 \sum_{k=0}^{N-1} \int_{\frac{k}{2N+1}}^{\frac{k+1}{2N+1}} |D_N(x)| dx \\
 &\geq 2 \sum_{k=0}^{N-1} \int_{\frac{k}{2N+1}}^{\frac{k+1}{2N+1}} \frac{|\sin((2N+1)\pi x)|}{\sin(\pi \frac{k+1}{2N+1})} dx \quad (\sin(\pi x) \text{ es creciente en } [0, 1/2)) \\
 &= 2 \sum_{k=0}^{N-1} \frac{1}{\sin(\pi \frac{k+1}{2N+1})} \underbrace{\int_k^{k+1} |\sin(\pi y)| \frac{dy}{2N+1}}_{=\frac{2}{\pi}} \quad (y = (2N+1)x) \\
 &= \frac{2}{2N+1} \sum_{k=0}^{N-1} \frac{1}{\sin(\pi \frac{k+1}{2N+1})} \frac{2}{\pi} \geq \frac{4}{2N+1} \frac{1}{\pi} \sum_{k=0}^{N-1} \frac{1}{\frac{k+1}{2N+1}} = \frac{4}{\pi} \sum_{k=0}^{N-1} \frac{1}{k+1} \\
 &\geq \frac{4}{\pi} \int_1^N \frac{1}{x} dx = \frac{4}{\pi} \log(N).
 \end{aligned}$$

Por tanto, podemos tomar $C = \frac{4}{\pi}$ y la desigualdad queda demostrada. ■

Observación 1. El `??` muestra que el núcleo de Dirichlet no es un núcleo de sumabilidad según la `nucleo-sumabilidad/Definición 1`, ya que no satisface la condición (2).

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